

Switch Case in C

A beginner's guide

1. What is switch?

A `switch` statement checks one value against several exact choices. It is useful for menus, month numbers, days, and calculator operations.

2. Basic syntax

```
switch (expression)
{
    case value1:
        // statements
        break;

    case value2:
        // statements
        break;

    default:
        // runs when no case matches
}
```

3. How it works

```
int number = 2;

switch (number)
{
    case 1:
        printf("One");
        break;

    case 2:
        printf("Two");
        break;

    default:
        printf("Invalid number");
}
```

The program finds the matching case, runs its statements, and stops at `break`.

4. Why do we use `break`?

`break` exits the switch. Without it, execution continues into the next cases. This is called **fall-through**.

```
switch (number)
{
    case 1:
        printf("One");
        break;

    case 2:
        printf("Two");
        break;
}
```

5. The default case

`default` runs when no case matches. It is similar to the final `else` in an `if-else` statement.

6. Month example

```
#include <stdio.h>

int main()
{
    int month;

    printf("Enter month number (1-12): ");
    scanf("%d", &month);

    switch (month)
    {
        case 4:
        case 6:
        case 9:
        case 11:
            printf("Number of days: 30");
            break;

        case 1:
        case 3:
```

```
    case 5:
    case 7:
    case 8:
    case 10:
    case 12:
        printf("Number of days: 31");
        break;

    case 2:
        printf("Number of days: 28 or 29");
        break;

    default:
        printf("Invalid month number");
}

return 0;
}
```

Cases 4, 6, 9, and 11 share the same code, so they are written together.

7. switch versus if-else

- Use `switch` for one value and exact choices.
- Use `if-else` for ranges or complex conditions, such as `marks >= 90`.

8. Important rules

- Case values must be constant values.
- Duplicate case values are not allowed.
- `break` prevents unwanted fall-through.
- `default` is optional but recommended.
- `switch` does not directly compare strings such as `"January"`.

Practice

Create a calculator menu: 1. Add, 2. Subtract, 3. Multiply, 4. Divide. Use `switch` to perform the selected operation.